

Stephen MacKenzie

Greater Seattle Area

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Summary

Experienced Software Development Engineer and Team Lead with over 20 years at Microsoft, specializing in C++ development, testing, and ISO standards compliance. Proven expertise in software development, quality assurance, and team leadership. Currently developing embedded libraries for safety-critical applications with a focus on automated frameworks and real-time systems. Passionate about contributing to advanced safety systems in the automotive and aerospace industries, and driving innovation through automated testing and development frameworks for embedded platforms.

Languages: C, C++, ARM Assembly, Rust (embedded development) Python

Embedded Platforms: ARM Cortex M series: STM32, Tiva C microcontrollers, Raspberry Pi

Development Tools: IAR Embedded Workbench, Keil uVision, STM32CubeMX (code generator), STM32CubeIDE, Code Composer Studio, PinMux, Resource Explorer, Driverlib, Visual Studio, VisualGDB, VSCode, Vim, NeoVim, Rust embedded crates, ST-Link, OpenOCD

Embedded Frameworks: STM32 HAL, CMSIS Programming, Legacy Std-Peripheral

Automated Testing Tools: CLI environment (gdb + stlink or openocd), STM32CubeMX integration for STM32 and PinMux/Resource Explorer for Tiva C series, CMSIS DSP Libraries for signal processing and testing

Software Engineering Skills: C++ and Rust Concurrency, Atomic Operations, Coroutines, Embedded Programming, Real-Time Systems, Data Integrity Techniques, Advanced Algorithms

Professional Experience

- **Self-Employed Performing Artist** Feb 2021 - Present
Bellevue, Washington, United States
 - Performed classical guitar concerts for charity.
 - Studied under Dr. Jon Yerby and achieved high-level performances.
- **Dell EMC Executive Consulting Architect** Aug 2018 - Jan 2019
Greater Seattle Area
 - Troubleshoot critical issues for Dell's acquisition of Isilon Systems.
 - Worked with C/C++, Assembly, Unix, Linux, and MySQL for high-level troubleshooting.
- **Director of a Non-Profit and Public Speaker** May 2014 - Aug 2018
Various Locations
 - Developed MLibs, an open-source suite of C and C++ libraries for embedded systems, focusing on automation and testing frameworks for microcontrollers.

- Delivered talks on embedded systems and containerization.
- Conducted charity work in Phoenix, Sacramento, Portland, and Tacoma.
- **Microsoft Software Engineer and Manager** 1993 - May 2014
Redmond, Washington, USA
 - Led the development and testing of C++ Libraries and backend tools.
 - Directed C++ security initiatives for Visual C++.
 - Contributed to ISO-standard-compliant C++ library development.

Projects

- **MLibs RTOS Framework:** Developed and maintained an open-source suite of C and C++ libraries for embedded systems, with a focus on automating testing and validation for microcontrollers.
- **x-cube-ai TensorFlow Lite Classification Project:** Implemented a text classification project using TensorFlow Lite on STM32, showcasing the integration of AI and embedded systems for real-time applications.
- **Classical Guitar Performances:** Performed benefit concerts, raising funds for charitable causes and promoting classical music in various communities.

Education

- **University of Washington** 2016 - 2017
Certification in Embedded and Real-Time Systems
- **Cornish College of the Arts**
Classical Guitar Studies

Skills

- Programming Languages: C++, C, Assembly, ARM Assembly, Python, Rust
- Embedded Systems: ARM, Cortex-M, IAR Embedded Workbench, GNU ARM Eclipse
- Development Tools: Visual Studio, Clang++, GCC, Make, CMake
- Software Development: Embedded Systems, Real-Time Systems, Safety-Critical Applications
- Standards Compliance: ISO C++ Standards, MISRA (de-emphasized here)

Honors & Awards

- Black Belt Award - Microsoft Corporation, Jan 1995
- Extra Mile Award - Microsoft Corporation, Mar 1994
- Washington State Programming Contest Winner, Mar 1992